Roll No. ....

TCE–501

B. TECH. (CIVIL ENGINEERING) (FIFTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2022

ENVIRONMENTAL ENGINEERING–I

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Write a short note of provision for "fire demand" in water supply. (CO1)

(b) List out the factors that are being considered in choosing between ground water and river water as a source of water supply of a town. (CO1)

(c) Describe the ways and means by which the water is being polluted by domestic and industries effluents. (CO1)

2. (a) Discuss briefly the common stresses produced in pipelines used for conveying water. List out the arrangements that are made in their design and construction to resist them. (CO2)

(b) Write a short note on corrosion of metal pipes. (CO2)

P. T. O.

(c) Give the reason for the following : (CO2)

(i) Hard rubber gasket is used in a flanged joint.

(ii) Copper pipes are used to convey hot water.

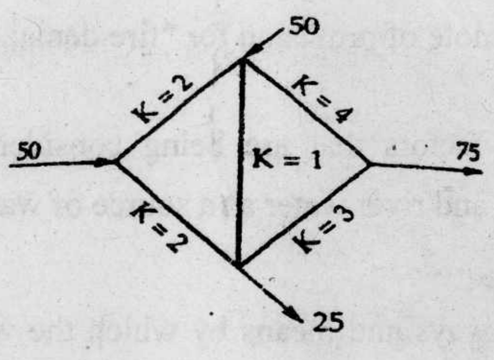
3. (a) Briefly explain the general methods of distribution of water employed in municipal water supply schemes. (CO3)

(b) Write short notes on the following : (CO3)

(i) Distribution of water to consumers

(ii) Causes of water waste, its detection and prevention

(c) Determine the distribution of flow in pipe network as shown in the figure. The head loss at h_L, May be assumed as KQ^n. The flow is turbulent, and pipes are rough. The value of capital K for each pipe is indicated in the figure. Use Hardy cross method. (CO3)



4. (a) The maximum daily demand at a water purification plant has been estimated as 12,000,000 litres per day. Design the dimension of a suitable sedimentation tank fitted with mechanical sludge removal arrangement for the raw supplies, assuming or retention period of 6 hours and velocity of flow as 20 cm per minute. (CO4)

(b) Draw the flowchart for water purification system. (CO4)

(c) Describe briefly the various coagulant used in water treatment. (CO4)

5. (a) List out the major air pollutants which causes impact on human health. (CO5, CO6)

(b) List out the major devices used to control the air and noise pollution. (CO5, CO6)

(c) Calculate the equivalent sound level when 2 sources of Sound having 60 decibel and 80 decibel are being monitored for a duration of 20 minutes and 25 minutes respectively. (CO5, CO6)

Roll No. ....

TCE–502

B. TECH. (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022

REINFORCED CEMENT CONCRETE—I

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

(v) Use of IS 456 : 2000 (Supplied by the Department) is allowed for use in the examination.

1. (a) Evaluate the characteristic strength of the following concrete strength investigation. Develop the Normal Distribution Curve and pass your comments on the result. (CO1)

Concrete Compressive Strength Observations								
	29	27.6	26.8	27.1	27.6	28.4	28.5	34
	30.2	32	30	29.3	30	28.4	29.1	32
	28.6	32	32.2	30.8	29	36	30	28
	34.6	32	29	26.9	29.5	33.4	31.5	31.5
	32	32	33	27.2	33	32.3	29.8	30.2

P. T. O.

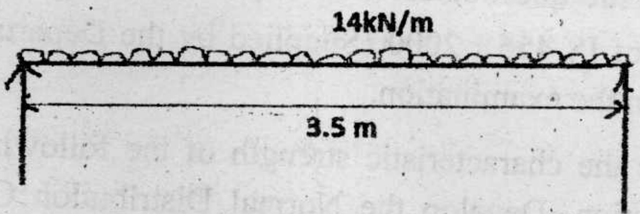
(b) Explain approach and the general sequence of design steps of LS method of design for flexure. (CO1)

(c) Present a comparative discussion in detail about the key aspects of working stress, ultimate load and limit state design philosophies. (CO2)

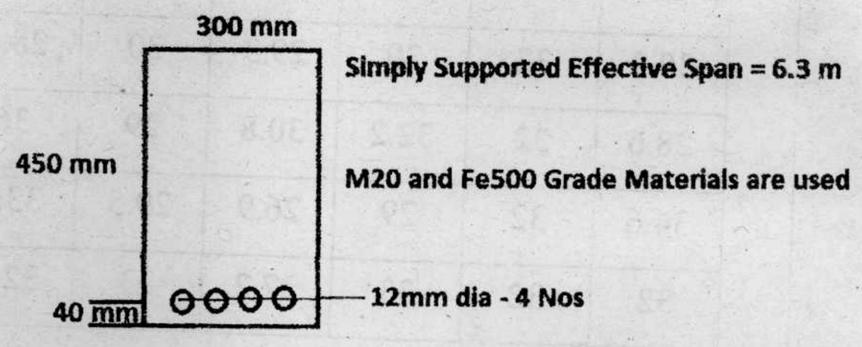
2. (a) Explain the concept and feasibility of under reinforced, over reinforced and balanced sections in the concrete flexure design philosophy. (CO2)

(b) Evaluate the Design Constants of M25 and Fe 500 Grade Steel in LSM or WSM. (CO2)

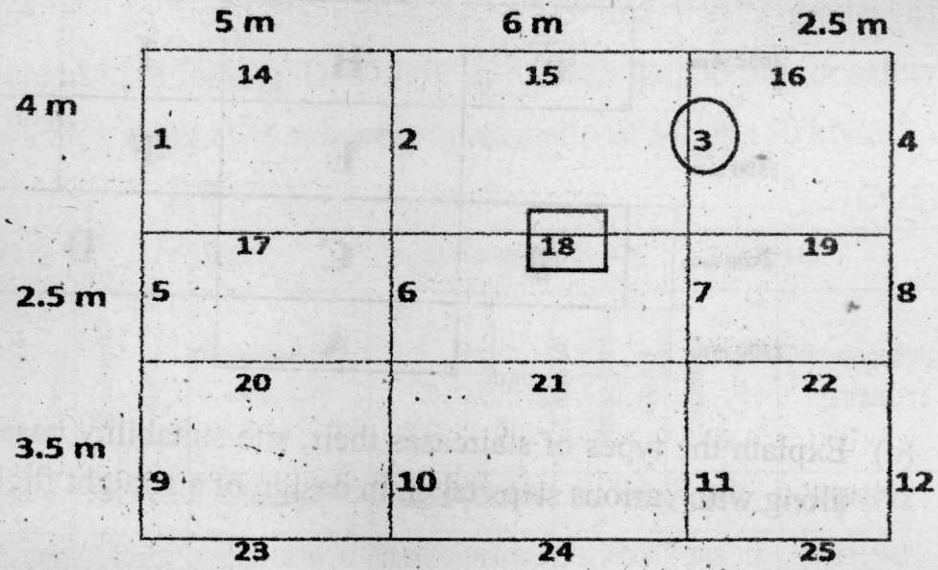
(c) Design a singly reinforced rectangular section in working stress method of design for the following beam. Use M15 and Fe415 grades of materials. (CO2)



3. (a) Decide the moment carrying capacity of a singly reinforced rectangular section of beam with the given data. (CO3)

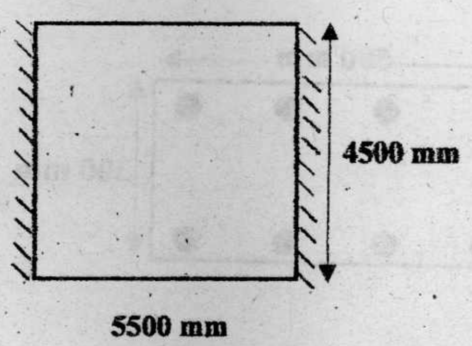


(b) Design a doubly reinforced rectangular beam section with web width restricted to 250 mm for the beam no. 18 with the given beam conditions. Live load on the slab is 3.0 kN/m^2, Floor Finish Load is 1 kN/m^2, Slab thickness is 120 mm casted in M20 Grade Concrete and Fe 415 Grade Steel with an effective cover of 20 mm. (CO3)

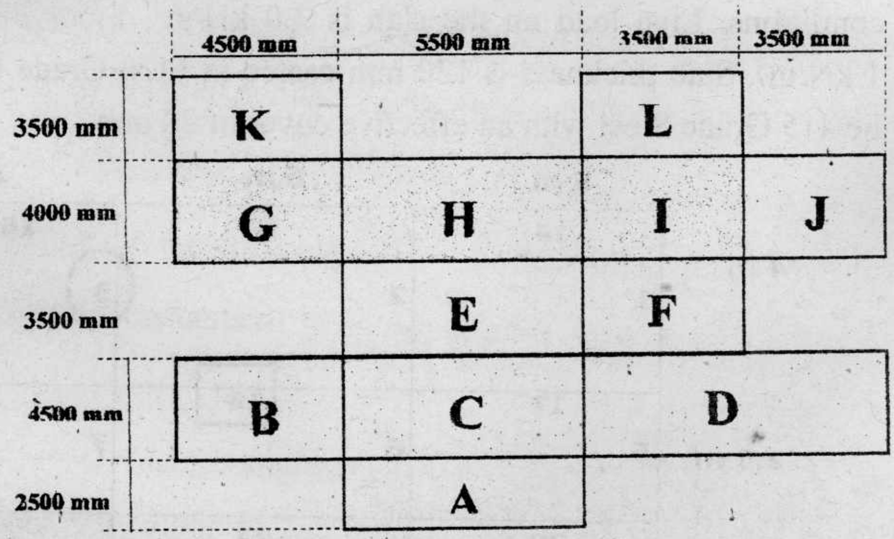


(c) Design a balanced singly reinforced T section, assuming that the concrete stress block completely falls in the flange portion only and also the width of web is restricted to 250 mm, using M20 Grade Concrete and Fe415 Grade Steel material for the beam no. 3 in the previous question. (CO3)

4. (a) Design and show the rough detailing of reinforcement for the given slab panel. M25 and Fe415 grade steel is being used. The slab panel is supposed to carry an overall working load of 7.5 kN/m^2 intensity over the entire area. (CO4).



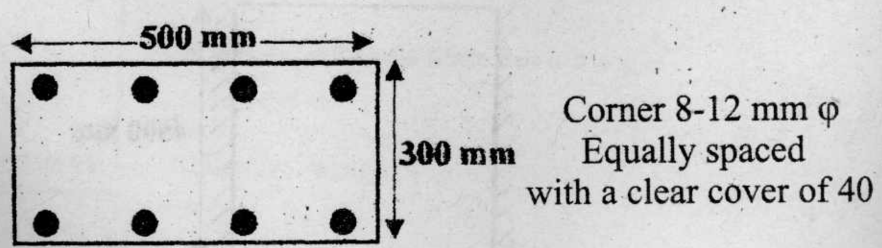
(b) Decide the Design Bending Moment of the given slab panel E. (CO4)



(c) Explain the types of staircases their, site suitability based on the flights along with various steps taken in design of a straight flight staircase. (CO4)

5. (a) Design a well distributed reinforcement for a square interior column of 300 × 300 mm for a floor of height 3.6 m, effectively restrained against rotation and lateral displacement to carry an overall working axial load of 1150 kN for which M30 and Fe415 grade steel are being used. (CO5)

(b) Determine the Ultimate Load Carrying capacity of the given Short Rectangular Column with the given cross section provided with horizontal lateral ties of 6 mm dia bar at a pitch of 160 mm. M25 and Fe415 grade steels are being used. (CO5)



(c) A square symmetric isolated footing of size 2.2 × 2.2 m is supporting a square column of size 500 × 500 mm size carrying a total axial factored load of 1500 kN. The overall depth of the footing is 700 mm, with a clear cover of 50 mm to the main reinforcement bars of 20 mm dia used at 150 mm c/c in both the directions. M25 and Fe415 grades of materials are used in footing. Determine and check the footing for safety against punching shear. The safe bearing capacity of soil is 150 kN/m^2.

(CO5)

Stress-Strain Values for			Fy =	500
K	Strain	Ultimate Stress-Strain Curve	Idealized Curve	Corresponding Strain
0.0000	0.0000	0.0000	0.00	0.000000
0.8000	0.0001	400.0000	347.83	0.001839
0.8500	0.0003	425.0000	369.57	0.002148
0.9000	0.0007	450.0000	391.30	0.002657
0.9500	0.0017	475.0000	413.04	0.003065
0.9750	0.0020	487.5000	423.91	0.004120
1.0000	0.0035	500.0000	434.78	0.005674

Stress-Strain Values for			Fy =	415
K	Strain	Ultimate Stress-Strain Curve	Idealized Curve	Corresponding Strain
0.0000	0.0000	0.0000	0.00	0.000000
0.8000	0.0001	332.0000	288.70	0.001543
0.8500	0.0003	352.7500	306.74	0.001834
0.9000	0.0007	373.5000	324.78	0.002324
0.9500	0.0017	394.2500	342.83	0.002714
0.9750	0.0020	404.6250	351.85	0.003759
1.0000	0.0035	415.0000	360.87	0.005304

Roll No. ....

TCE–503

B. TECH. (FIFTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2022

GEOTECHNICAL ENGG.—II

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks.

1. (a) Mention (only) different methods of analyzing the stability of the finite slope. Provide detailed discussion on estimating factor of safety of finite slope using any "Limit equilibrium approach". (CO1)

(b) Discuss in detail the three conditions for which the stability analysis of an earth dam is to be carried out. (CO1)

(c) A cutting is to be made in cohesive-frictional soil for which the cohesion is 55 kN/m^2 and ϕ =30°. The density of the soil is 18 kN/m^3. Find the safe depth of unsupported cut, if the factor of safety is to be 1.5. (CO1)

P. T. O.

2. (a) Discuss the following approximate distribution methods for loaded area : Equivalent point load method and two to one (2 : 1) method. (CO2)

(b) Qualitatively and jointly present the stress distribution diagram due to point load for vertical stress isobar diagram at a distance r from the point load and vertical stress distribution on a horizontal plane z depth below the ground surface. (CO2)

(c) A circular ring-type foundation for a water tank exerts on the soil a uniformly distributed pressure of 200 kPa. The outer diameter of the ring foundation is 5.5 meter and thickness of the ring is 1.5 m. Estimate the amount of vertical stress produced at a depth of 3.0 m below the center of the ring footing. Use following equation for calcualting the stress due to uniformly loaded (q) circular area.

$$\sigma_z = q\left[1 - \frac{1}{1 + \left(\frac{R}{z}\right)^2}\right]^{3/2}$$

R = radius of the circular area, z = depth below the center of the loaded area. (CO2)

3. (a) List down different methods of field test. How you can estimate the load carrying capacity of the foundation soil using any field method ? (CO3)

(b) A raft of size 4 m-square carries a load of 200 kN/m². Determine the vertical stress increment at a point 4 m below the center of the loaded area using equivalent point load method by dividing the area into four equal parts the load from each of which is assumed to act through its center. (CO3)

(c) A plate load test was conducted on a uniform deposit of sand and the following data were obtained : (CO3)

Pressure (kN/m^2)	Settlement (mm)
50	1.5
100	2.0
200	4.0
300	7.5
400	12.5
500	20.0
600	40.0

The size of the plate used was 500 mm × 500 mm and that of the pit 4m × 4m × 1.5 m.

(i) Plot the pressure-settlement curve on a scale and determine the failure stress.

(ii) A square footing, $2m \times 2m$, is to be founded at 1.5 m depth in this soil. Determine the allowable bearing pressure settlement of 40 mm.

4. (a) Define the term "Negative Skin Friction" and explain how it affects the load carrying capacity of the single pile. Derive expression for estimating the load carrying capacity of single pile installed in saturated clay in which top few meters area subjected to negative skin friction.

(CO4)

(b) Classify the pile foundation based on (i) shape (ii) material used (iii) method of installation. How to estimate the load carrying capacity of single pile intalled in sand. Develop the expression. (CO4)

(c) A group of nine piles ware arranged in square pattern and installed in cohesive soil. The diameters of the piles are 300 mm and their c/c spacing is 900 mm. Length of each pile is 15 m. Assume all piles are friction piles. The undrained cohesion value of the foundation soil is 150 kPa. Estimate the ultimate load carrying capacity of the pile group. Take adhesion factor $\alpha = 0.85$. (CO4)

5. (a) Discuss in detail field test "SPT" with advantages and disadvantages. Also, discuss various corrections applied to raw SPT values are as per Indian standard code of practice. (CO5)

(b) Mention (only) different types of sample used in the field. Define the terms Area ratio, Recovery ratio and Clearance. (CO5)

(c) Write short notes on the following ground improvement techniques : Pre-loading and Compaction. (CO5)

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Roll No. ....

TCE–504

B. TECH. (CIVIL ENGINEERING) (FIFTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2022

WATER RESOURCES ENGINEERING—I

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

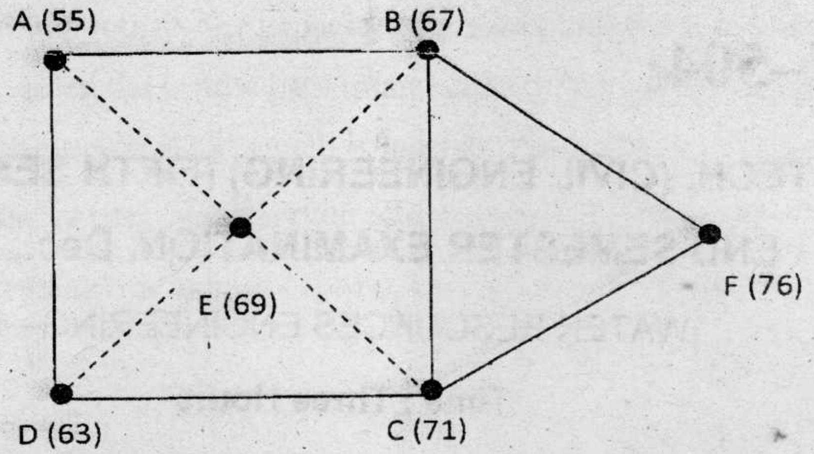
1. (a) Differentiate between a recording and non-recoding rain gauge, and list the considerations for the selection of a site for a rain gauge station. (CO1)

(b) What are the considerations in the design of rain gauge network ? What are the norms of rain gauge density ? How do you determine optimum number of rain gauges. (CO1)

(c) Find the mean precipitation for the area sketched below by Theisson method. The area is composed of a square plus an equilateral triangular

P. T. O.

plot of side 10 km. Rainfall readings in mm at the various stations are given. Check the same by arithmetic average method. (CO1)



2. (a) Describe the procedure of deriving the ordinates of UH from the given ordinates of flood hydrograph. Discuss the various methods of base flow separation. (CO2)

(b) The ordinates of a 2-hr UH for aparticular basin are given below. Derive the ordinates of : (CO2)

(i) S-curve hydrograph

(ii) Area of basin

Time (hr)	2-hr UHGO
0	0
1	100
2	250
3	200
4	100
5	50
6	0

(c) Flood frequency computations for a flashy river at a point 50 km upstream of a bund site indicated the following : (CO2)

Return period (years)	Peak flood (m^3/s)
50	20600
100	22150

Estimate the flood magnitude in the river with a return period of 500 year through use of Gumbel's method.

3. (a) Establish a relation between Duty of water, Delta and Base period. (CO3, CO6)

(b) A 30 cm well completely penetrates an unconfined aquifer of depth 40 m. After a long period of pumping at a steady rate of 1500 lpm, the drawdown in two observation wells 25 m and 75 m from the pumping well were found to be 3.5 m and 2.0 m, respectively. Determine the transmissibility of the aquifer. What is the drawdown at the pumping well ? (CO3, CO6)

(c) The base period, intensity of irrigation, and duty of water for various crops under the canal system are given. Determine the reservoir capacity if the culturable commanded area is 40,000 hectares, canal losses are 25%, and reservoir losses are 15% : (CO3, CO6)

Crops	Base period	Duty of the water at the field days hec/cumec	Intensity of irrigation %
Wheat	120	1800	20
Sugarcane	360	1700	20
Cotton	180	1400	10
Rice	120	800	15
Vegetables	120	700	15

4. (a) What are the factors that affect the pattern of sediment deposition in a reservoir ? (CO4)

(b) How would you fix the capacity of a dam reservoir at a particular river site, provided the inflow pattern and demand pattern are known. Explain the mass curve method which is used for this purpose . (CO4)

(c) A contour survey of a reservoir site gives the following data :

Contour value	Area
At 100 m contour	6.0 hact
At 110 m contour	18.1 hact
At 120 m contour	34.0 hact

The capacity of reservoir up to 100 m elevation is found to be 14.1 hact-m. Determine the general equation for the area elevation curve and capacity elevation curve. Also determine the reservoir capacity at RL 125 m. (CO4)

5. (a) Briefly describe and discuss the various methods of lining irrigation canals. Give a cross-section of a lined canal. (CO5)

(b) Design a regime channel for a discharge of 35 m^3/s with silt factor of 0.9 by Lacey's theory, taking side slopes as 1 H : 2 V. (CO5)

(c) What are the different types of cross drainage works ? State the conditions under which each one is adopted. Sketch any one of them. (CO5)

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TCE–505

B. TECH. (FIFTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2022

STRUCTURAL ANALYSIS—II

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

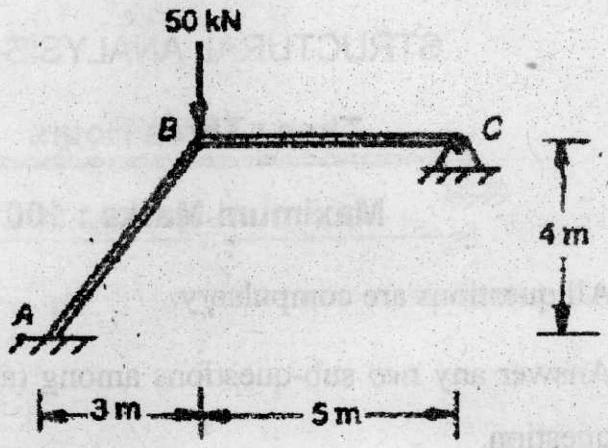
1. (a) Find the horizontal thrust in a two-hinged semi-circular arch of radius 'R' subjected to a vertical concentrated load 'W' on a section that is inclined at an angle 'α' with the positive x-axis. (CO1)

(b) A suspension bridge of 100 m span has a three hinged stiffening girder supported by cables having a central dip of 10 m. The left half of the span is loaded with a uniformly distributed load of 30 kN/m. Determine the reactions and draw the bending moment diagram. (CO1)

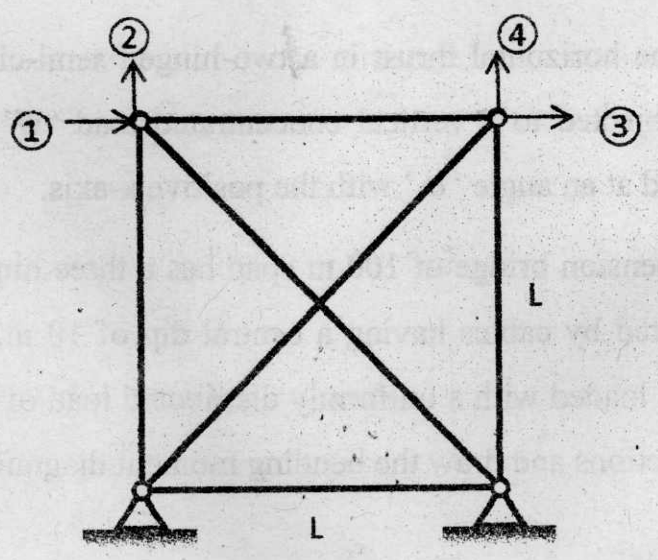
P. T. O.

(c) A symmetrical two-hinged parabolic arch has a span of 20 m and a central rise of 4 m. Find the horizontal thrust at the support if the central 10 m span carries a udl of 24 kN/m measured horizontally. Assume secant variation of EI. (CO1)

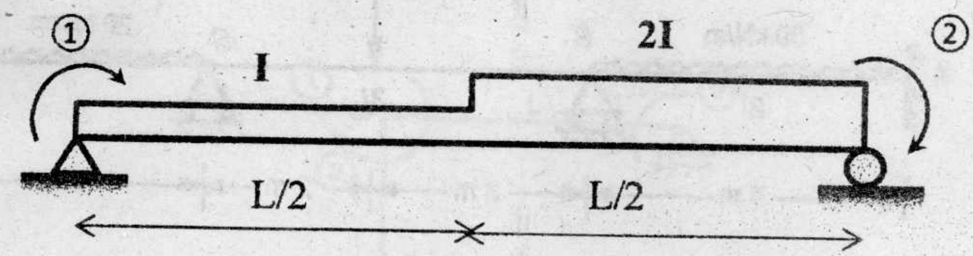
2. (a) Analyze the plane frame shown below by force method : (CO2)



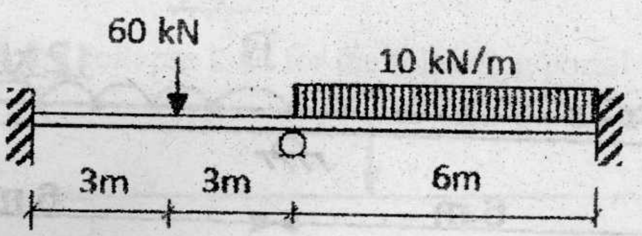
(b) Determine the flexibility matrix for the plane truss shown below : (CO2)



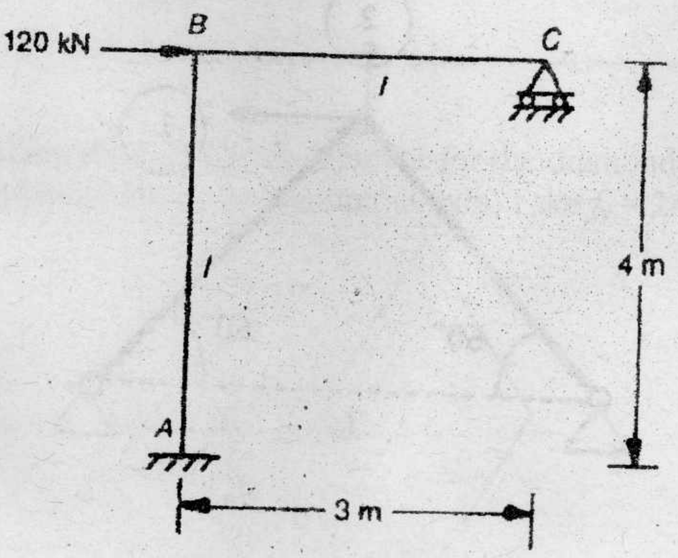
(c) Obtain the flexibility matrix for the non-prismatic beam shown below : (CO2)



3\. (a) Find all the support moments using moment distribution method for the continuous beam shown below : (CO3)

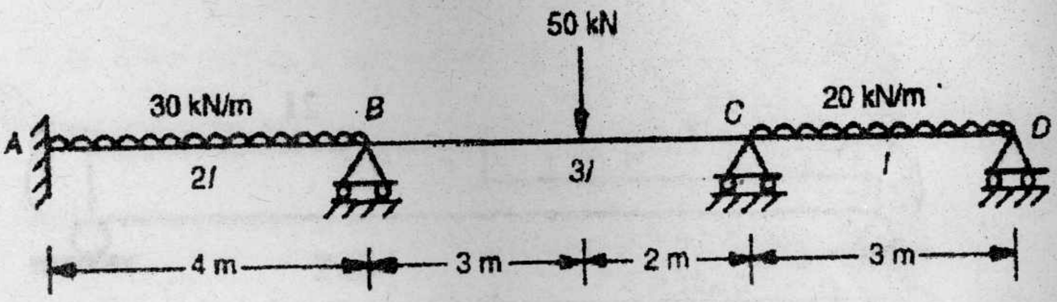


(b) Analyse the following rigid jointed plane frame using moment distribution or slope deflection method. The horizontal load at joint B is 120 kN. (CO3)

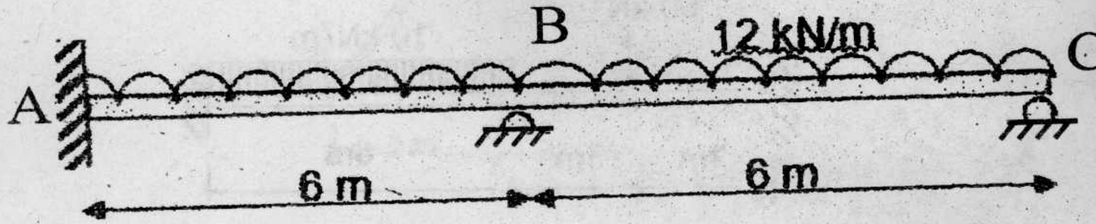


P. T. O.

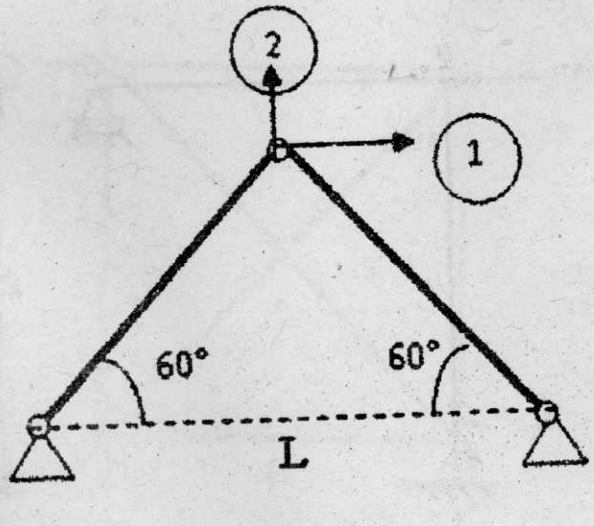
(c) Analyse the following beam using moment distribution method : (CO3



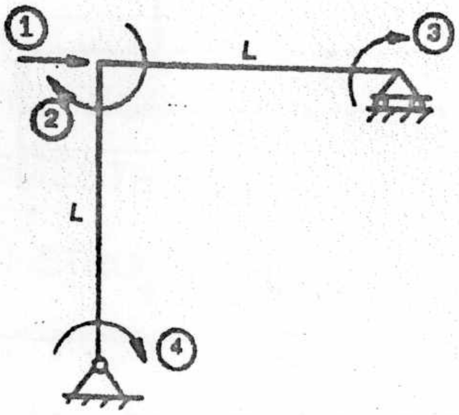
4. (a) Determine the rotations at supports B and C of the continuous beam shown below : (CO4)



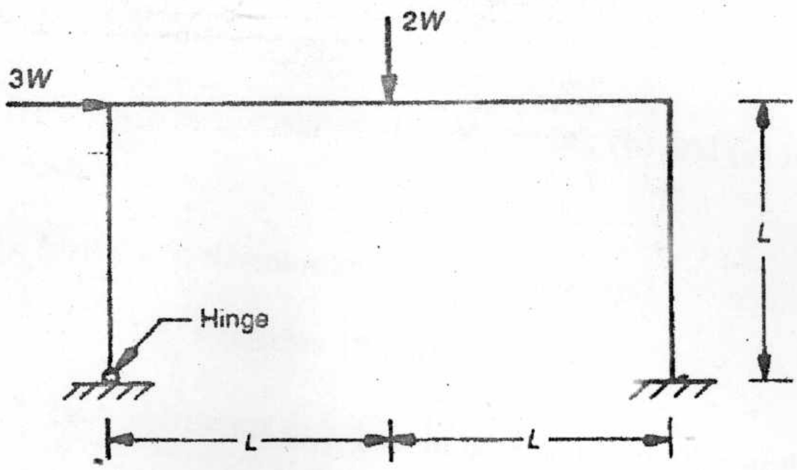
(b) Obtain the stiffness matrix for the following truss structure : (CO4)



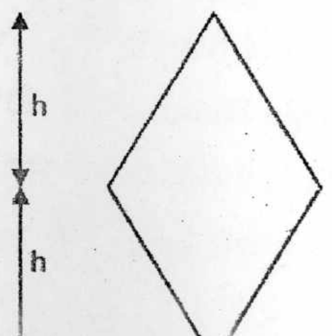
(c) Generate the stiffness matrix of the rigid jointed frame shown below : (CO4)



5. (a) Determine the collapse load for the following portal frame : (CO5)

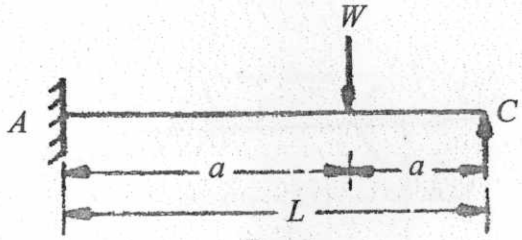


(b) Determine the full plastic moment for the diamond shaped beam section shown below about the horizontal axis. Take $f_y = 250$ N/mm^2. (CO5)



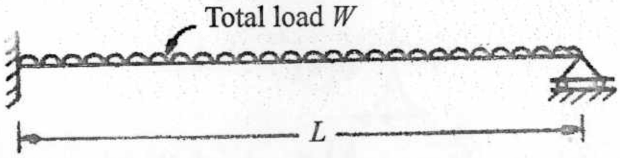
(c) Determine the collapse load for the following structures : (CO5)

(i)



Beam

(ii)



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